

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and zero-copy cross-process message passing
- Profiled settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven **TypeScript/React** observer system across **25K+ lines** of Chromium at **95% test coverage**; design documents reviewed and adopted into production branch by senior Chrome engineers

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Diagnosed DynamoDB hot-partition failure at **9,000+ req/sec**; redesigned partition key strategy and access patterns — **30% read throughput improvement** under sustained high-concurrency traffic
- Designed idempotent payment APIs with Redis distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS — selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**; diagnosed N+1 query bottleneck, redesigned **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

PERSONAL PROJECTS

TiMoto AI – Systems & Performance Platform | C++, gRPC, Python, AWS, K8s

timoto.ai

- **Sole engineer** — designed, built, and operate distributed backend platform: **gRPC** inter-service layer, Python/Django REST backend, PostgreSQL, and multi-AZ **AWS** infrastructure with Terraform IaC
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause to circular resource acquisition; redesigned call sequencing with explicit lock ordering — **100% success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with circuit breaker pattern and health-check-triggered auto-rollback — **99.9% uptime, 44% cost reduction** to \$40–60/month under production traffic
- Integrated vLLM inference engine; resolved KV cache memory fragmentation via **PagedAttention** — zero OOM failures under concurrent load through continuous batching and memory-aware scheduling

Pulumi – Open Source Contributor | Go, TypeScript, IaC

24.4K★ github.com/pulumi/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

TECHNICAL SKILLS

Languages: C++, Python, Go, Rust, TypeScript, Java, SQL, Bash

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Raft/Paxos, Fault tolerance, Linearizability

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS), Kubernetes, Docker, Terraform, CI/CD, GitHub Actions

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, MySQL, Redis, DynamoDB

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, PyTorch, Ray